

Paper 1 - Set 10 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Binomial expansion
2	B	Exponentials and logarithms
3	F	Angles, polygons, congruence and vectors
4	F	Sequences and series
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6	D	Integration
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Worked solutions

Question 1 BIN-07 · Binomial expansion · spec MM2.4

Expand $(1 - 2x)^4$ and hence find the coefficient of x^2 in $(1 + 3x)(1 - 2x)^4$.

Answer: D

$(1 - 2x)^4 = 1 - 8x + 24x^2 - 32x^3 + 16x^4$. Coefficient of x^2 in the product: $1 \cdot 24 + 3 \cdot (-8) = 0$.

Full method - how and why

A product of a binomial expansion with a linear factor. Only two products can give x^2 , so collect just those.

1 Expand $(1 - 2x)^4$: coefficients $\binom{4}{r}(-2)^r$ give $1 - 8x + 24x^2 - 32x^3 + 16x^4$.

why: $\binom{4}{1}(-2) = -8$ and $\binom{4}{2}(-2)^2 = 6 \times 4 = 24$; remember to raise the -2 to the power as well.

2 To make x^2 in the product, pair $1 \times 24x^2$ and $3x \times (-8x)$.

why: Degrees must add to 2: (0, 2) and (1, 1) are the only options.

3 Add: $24 + (-24) = 0$.

why: The two contributions cancel exactly - the answer is 0, not 'no term'.

Where it goes wrong

- Taking the x^2 coefficient of $(1 - 2x)^4$ as +24 but forgetting the $3x \times (-8x)$ contribution, giving 24.
- Sign error making $(-2)^2 = -4$.

Takeaway For a coefficient in a product, list the degree pairs that add to the target and sum their contributions.

Question 2 LOG-18 · Exponentials and logarithms · spec MM5.2

If $\log_a b = 2$ and $\log_b c = 3$, find $\log_a c$.

Answer: B

$b = a^2$ and $c = b^3 = a^6$, so $\log_a c = 6$.

Full method - how and why

A chain of logarithms. Convert each statement to exponential form and substitute.

1 $\log_a b = 2$ means $b = a^2$.

why: $\log_a b = c \Leftrightarrow a^c = b$.

2 $\log_b c = 3$ means $c = b^3$.

why: Same conversion with the next pair.

3 Substitute: $c = (a^2)^3 = a^6$.

why: Powers of powers multiply.

4 Therefore $\log_a c = 6$.

why: Read the exponent straight off. (In general $\log_a c = \log_a b \times \log_b c = 2 \times 3$.)

Where it goes wrong

- Adding the values to get 5.
- Reading $(a^2)^3$ as a^5 .

Takeaway Logarithms chain by multiplication: $\log_a c = \log_a b \times \log_b c$.

Question 3 GEO-01 · Angles, polygons, congruence and vectors · spec M5.2

The interior angle of a regular polygon is 156° . How many sides does it have?

Answer: F

Exterior angle 24° ; $360/24 = 15$ sides.

Full method - how and why

Work with the **exterior** angle: they always sum to 360° , which makes the arithmetic trivial.

- 1 Exterior angle $= 180^\circ - 156^\circ = 24^\circ$.

why: Interior and exterior angles at a vertex lie on a straight line.

- 2 The exterior angles of any polygon sum to 360° , and for a regular polygon they are all equal.

why: Walking once around the shape turns you through a full circle.

- 3 Number of sides $= \frac{360}{24} = 15$.

why: Divide the total turn by the turn per vertex.

Where it goes wrong

- Using $\frac{360}{156}$ or $\frac{156}{24}$.
- Using the interior-angle-sum formula and then having to solve $\frac{180(n-2)}{n} = 156$ - correct but slower.

Takeaway For regular polygons, go via the exterior angle: $n = \frac{360}{\text{exterior}}$.

Question 4 SEQ-10 · Sequences and series · spec MM2.3

A geometric series has first term 2 and common ratio $r > 0$. The sum of the first three terms is 14. Find the sum of the first five terms.

Answer: F

$$2 + 2r + 2r^2 = 14 \Rightarrow r^2 + r - 6 = 0 \Rightarrow r = 2. S_5 = 2(2^5 - 1) = 62.$$

Full method - how and why

The first three terms give a quadratic in r ; the positivity condition picks one root.

- 1 Write the sum of three terms: $2 + 2r + 2r^2 = 14$.

why: Terms are a , ar , ar^2 with $a = 2$.

- 2 Divide by 2 and rearrange: $r^2 + r - 6 = 0$.

why: Dividing first keeps the numbers small.

- 3 Factorise: $(r + 3)(r - 2) = 0$, so $r = 2$ or $r = -3$. The condition $r > 0$ gives $r = 2$.

why: The question's restriction exists precisely to discard the negative root.

- 4 Sum five terms: $S_5 = 2 \frac{2^5 - 1}{2 - 1} = 2 \times 31 = 62$.

why: $2^5 = 32$, so the bracket is 31.

Where it goes wrong

- Using $r = -3$ and getting $S_5 = 2 \times \frac{-243 - 1}{-4} = 122$.
- Adding two more terms by hand and slipping: $2 + 4 + 8 + 16 + 32 = 62$ is a good check.

Takeaway Conditions like ' $r > 0$ ' are there to select between roots - apply them before continuing.

Question 5 IND-14 · Indices and surds · spec MM1.2

The number $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$ is equal to

Answer: A

$$\sqrt{50} = 5\sqrt{2}, \sqrt{18} = 3\sqrt{2}, \sqrt{8} = 2\sqrt{2}. \text{ So } \frac{8\sqrt{2}}{2\sqrt{2}} = 4.$$

Full method - how and why

A quotient of surds that collapses completely. Simplify each root first and the $\sqrt{2}$'s cancel.

1 Simplify: $\sqrt{50} = 5\sqrt{2}, \sqrt{18} = 3\sqrt{2}, \sqrt{8} = 2\sqrt{2}$.

why: Every number is (square) $\times 2$, so every root is a multiple of $\sqrt{2}$ - which is why the fraction will be rational.

2 Add the numerator: $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$.

why: Like terms again - $\sqrt{2}$ behaves as a common factor.

3 Divide: $\frac{8\sqrt{2}}{2\sqrt{2}} = 4$.

why: The $\sqrt{2}$ cancels top and bottom, leaving plain arithmetic.

Where it goes wrong

- Working out $\sqrt{50} + \sqrt{18} = \sqrt{68}$, which is wrong and leads nowhere.
- Cancelling only the numbers and leaving a stray $\sqrt{2}$, giving $4\sqrt{2}$.

Takeaway Simplify every surd to the form $k\sqrt{m}$ before adding or dividing; common surd factors then cancel visibly.

Question 6 INT-20 · Integration · spec MM7.2

Evaluate $\int_{-1}^2 |x| dx$.

Answer: D

Two triangles: $\frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 2 \cdot 2 = \frac{1}{2} + 2 = \frac{5}{2}$.

Full method - how and why

A modulus integrand. Split at the corner, or - faster - read the areas off as triangles.

1 Note $|x|$ changes formula at $x = 0$, which is inside $[-1, 2]$.

why: Split at every point where the modulus switches branch.

2 On $[-1, 0]$: $|x| = -x$, and the graph is a triangle with base 1 and height 1, area $\frac{1}{2}$.

why: Sketching $y = |x|$ makes the integral a pair of triangles - no antiderivative needed.

3 On $[0, 2]$: $|x| = x$, a triangle with base 2 and height 2, area 2.

why: $\frac{1}{2} \times 2 \times 2 = 2$.

4 Add: $\frac{1}{2} + 2 = \frac{5}{2}$.

why: Both pieces are above the axis, so both count positively.

Where it goes wrong

- Integrating x straight through and getting $\frac{4}{2} - \frac{1}{2} = \frac{3}{2}$ - an offered option.
- Treating the left piece as negative.

Takeaway $|x|$ is never negative: split at the corner, and use triangle areas when the graph is piecewise linear.

Question 7 POL-02 · Polynomials, factor and remainder theorems · spec MM1.6

$(x + 2)$ is a factor of $2x^3 + ax^2 - 5x - 6$. Find the value of a .

Answer: F

Factor Theorem: $f(-2) = -16 + 4a + 10 - 6 = 4a - 12 = 0$, so $a = 3$.

Full method - how and why

'Is a factor' means remainder zero, so the Factor Theorem turns the statement into a single equation for a .

- 1 $(x + 2)$ vanishes at $x = -2$.

why: Write $x + 2 = x - (-2)$ if the sign is confusing; the root is the value that makes the bracket zero.

- 2 Substitute: $f(-2) = 2(-8) + a(4) - 5(-2) - 6 = -16 + 4a + 10 - 6$.

why: Careful with the odd power: $(-2)^3 = -8$, while $(-2)^2 = +4$.

- 3 Simplify: $4a - 12$.

why: $-16 + 10 - 6 = -12$.

- 4 Set to zero (factor $\Rightarrow$ remainder zero): $4a = 12$, so $a = 3$.

why: That is precisely the Factor Theorem.

Where it goes wrong

- Using $x = 2$ instead of $x = -2$, which gives $a = -3$ - an offered option.
- Sign slips on $(-2)^3$ and $-5(-2)$; write each term out separately.

Takeaway Factor $(x + k)$ means substitute $x = -k$ and set the result to zero.

Question 8 MEN-14 · Pythagoras, mensuration and 3D · spec M5.16

A cylindrical tin of radius 4 cm and height 10 cm is full of water. The water is poured into an empty cylindrical jug of radius 5 cm. How deep is the water in the jug?

Answer: F

$$\pi \cdot 16 \cdot 10 = \pi \cdot 25 \cdot h \Rightarrow h = \frac{160}{25} = 6.4.$$

Full method - how and why

Conservation of volume: the water keeps its volume, so equate the two cylinder volumes.

- 1 Volume in the tin: $\pi r^2 h = \pi(16)(10) = 160\pi$.

why: $r = 4$ so $r^2 = 16$.

- 2 In the jug the same volume occupies $\pi(25)h$.

why: Radius 5, unknown depth h ; the π will cancel.

- 3 Equate and solve: $160\pi = 25\pi h \Rightarrow h = \frac{160}{25} = 6.4$ cm.

why: The wider jug means a shallower depth - a good sanity check.

Where it goes wrong

- Using the radii rather than their squares, giving 8 cm - an offered option.
- Expecting the depth to increase because the jug is 'bigger'.

Takeaway Pouring conserves volume: equate $\pi r_1^2 h_1 = \pi r_2^2 h_2$; a wider container gives a shallower depth.

Question 9 SEQ-09 · Sequences and series · spec MM2.2

How many terms of the arithmetic series $3 + 7 + 11 + \dots$ are needed for the sum to first exceed 500?

Answer: B

$$S_n = \frac{n}{2}(6 + 4(n - 1)) = n(2n + 1). \quad n = 15: 465; \quad n = 16: 528 > 500. \text{ So } 16.$$

Full method - how and why

An inequality on a sum. Form S_n , then find the first integer n that works - by trial, since n must be a whole number.

- 1 Identify $a = 3$, $d = 4$ and form the sum: $S_n = \frac{n}{2}[6 + 4(n - 1)] = \frac{n}{2}(4n + 2) = n(2n + 1)$.

why: Simplifying to $n(2n + 1)$ makes the trial values much easier to compute.

- 2 Test $n = 15$: $15 \times 31 = 465$, which is less than 500.

why: Estimate first: $2n^2 \approx 500$ gives $n \approx 15.8$, so start near there.

- 3 Test $n = 16$: $16 \times 33 = 528 > 500$ ✓.

why: The sum is increasing, so the first n that exceeds 500 is the answer.

- 4 Answer: 16 terms.

why: Check the one below to be sure it fails - that is what 'first exceeds' demands.

Where it goes wrong

- Solving $2n^2 + n = 500$ and rounding down to 15.
- Forgetting that n must be a positive integer and quoting a decimal.

Takeaway For 'first exceeds' questions, solve approximately then test the two neighbouring integers.

Question 10 BIN-05 · Binomial expansion · spec MM2.4

Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.

Answer: A

$$\binom{10}{3} = \binom{10}{7}, \text{ so the first two cancel; } \binom{9}{2} = 36.$$

Full method - how and why

A short calculation made trivial by the symmetry of binomial coefficients.

- 1 Use the symmetry $\binom{n}{r} = \binom{n}{n-r}$: since $3 + 7 = 10$, $\binom{10}{3} = \binom{10}{7}$.

why: Choosing 3 objects from 10 is the same as choosing which 7 to leave behind.

- 2 So the first two terms cancel, leaving $\binom{9}{2}$.

why: Spotting the cancellation avoids computing 120 twice.

- 3 Evaluate $\binom{9}{2} = \frac{9 \times 8}{2} = 36$.

why: $\binom{n}{2} = \frac{n(n-1)}{2}$.

Where it goes wrong

- Computing all three coefficients and mis-adding, e.g. $120 - 120 + 36$ done as 156.
- Thinking $\binom{10}{3} - \binom{10}{7}$ is $\binom{10}{-4}$ or similar.

Takeaway $\binom{n}{r} = \binom{n}{n-r}$: look for pairs whose lower entries add to n .

Question 11 TRG-16 · Trigonometric functions and equations · spec MM4.6

Given that $\sin x + \cos x = 0$ and $0 \leq x < 2\pi$, find the sum of all possible values of x .

Answer: B

$$\tan x = -1: x = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}. \text{ Sum} = \frac{10\pi}{4} = \frac{5\pi}{2}.$$

Full method - how and why

Convert a sum of sin and cos into a single tan equation by dividing.

- 1 Rearrange: $\sin x = -\cos x$.
why: Move one term across.
- 2 Divide by $\cos x$: $\tan x = -1$. (Check $\cos x = 0$ separately: then $\sin x = 0$ too, impossible.)
why: Dividing by $\cos x$ is safe only after ruling out $\cos x = 0$ - and here that case gives a contradiction.
- 3 Solve on $[0, 2\pi)$: $\tan x = -1$ at $x = \frac{3\pi}{4}$ and $x = \frac{3\pi}{4} + \pi = \frac{7\pi}{4}$.
why: Reference angle $\frac{\pi}{4}$, negative tangent in quadrants 2 and 4.
- 4 Sum: $\frac{3\pi}{4} + \frac{7\pi}{4} = \frac{10\pi}{4} = \frac{5\pi}{2}$.
why: Add over the common denominator and simplify.

Where it goes wrong

- Finding only one solution and answering $\frac{3\pi}{4}$ - an offered option.
- Forgetting to justify dividing by $\cos x$ (here it is fine, but the habit matters).

Takeaway $a \sin x + b \cos x = 0$ becomes $\tan x = -\frac{b}{a}$; check the $\cos x = 0$ case before dividing.

Question 12 NUM-16 · Number, units and estimation · spec M1.1

Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume 3.6 m^3 ? ($1 \text{ m}^3 = 1000$ litres.)

Answer: B

3600 litres at 1.2 litres/s takes $3000 \text{ s} = 50$ minutes.

Full method - how and why

A rate problem with a unit conversion. Convert the volume to litres, divide by the rate, then convert the time.

- 1 Convert the volume: $3.6 \text{ m}^3 = 3.6 \times 1000 = 3600$ litres.
why: The conversion is given in the question.
- 2 Divide by the rate: $\frac{3600}{1.2} = 3000$ seconds.
why: Volume divided by volume-per-second gives seconds.
- 3 Convert to minutes: $\frac{3000}{60} = 50$ minutes.
why: The options are in minutes, so finish the conversion.

Where it goes wrong

- Leaving the answer as 3000 seconds and picking 500 seconds by a slip.
- Dividing by 1.2 as if it were 12, giving 5 minutes.

Takeaway Make the units match before dividing, and convert the answer into the units the options use.

Question 13 GRA-02 · Graphs of functions · spec MM8.7

The graphs of $y = x^2 - 2x$ and $y = k$ meet at exactly two points. Find the complete set of values of k .

Answer: A

$y = (x - 1)^2 - 1$ has minimum -1 ; a horizontal line above the minimum meets the parabola twice.

Full method - how and why

A horizontal line meeting a parabola twice: the line must be above the minimum.

- 1 Complete the square: $x^2 - 2x = (x - 1)^2 - 1$, so the minimum value is -1 at $x = 1$.

why: The turning point governs how many times a horizontal line can cut the curve.

- 2 A line $y = k$ meets the parabola twice when k is **strictly above** the minimum.

why: At $k = -1$ the line is tangent at the vertex: exactly one point. Below it: none.

- 3 So $k > -1$.

why: Strict inequality, because equality gives only one intersection.

Where it goes wrong

- Including $k = -1$ and choosing $k \geq -1$ - offered as a distractor.
- Using the y -intercept 0 instead of the minimum value.

Takeaway Count intersections with a horizontal line by sliding it against the turning point; equality gives tangency.

Question 14 ALG-14 · GCSE algebra · spec M4.16

The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

Answer: C

$$x^2 - y^2 = (x - y)(x + y) = 12(x - y) = 48 \Rightarrow x - y = 4. \text{ So } x = 8, y = 4.$$

Full method - how and why

A sum and a difference of squares. Factorising the difference of squares links them without solving for both numbers.

- 1 Factorise: $x^2 - y^2 = (x - y)(x + y)$.

why: This is the identity that connects the two given facts.

- 2 Substitute the known sum: $48 = (x - y)(12)$, so $x - y = 4$.

why: The sum $x + y = 12$ is given directly.

- 3 Solve the pair $x + y = 12$, $x - y = 4$: adding gives $2x = 16$, so $x = 8$ (and $y = 4$).

why: Adding the two equations eliminates y .

- 4 The larger number is 8.

why: Check: $64 - 16 = 48$ ✓.

Where it goes wrong

- Answering 4 (the smaller number or the difference).
- Trying to solve $x^2 - y^2 = 48$ by trial.

Takeaway $x^2 - y^2 = (x - y)(x + y)$ converts a difference of squares into a linear pair.

Question 15 LOG-01 · Exponentials and logarithms · spec MM5.2

Find the value of $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$.

Answer: A

$$\log_2 \left(\frac{40}{5} \cdot \frac{1}{4} \right) = \log_2 2 = 1.$$

Full method - how and why

Combine everything into a single logarithm using the laws, then evaluate.

1 Combine the first two: $\log_2 40 - \log_2 5 = \log_2 \frac{40}{5} = \log_2 8$.

why: A difference of logs (same base) is the log of the quotient.

2 Add the third: $\log_2 8 + \log_2 \frac{1}{4} = \log_2 \left(8 \times \frac{1}{4} \right) = \log_2 2$.

why: A sum of logs is the log of the product. Note $\log_2 \frac{1}{4}$ is negative, so this step reduces the total.

3 Evaluate: $\log_2 2 = 1$.

why: $\log_a a = 1$ for any valid base.

Where it goes wrong

- Treating $\log_2 \frac{1}{4}$ as $+2$ rather than -2 and answering 5.
- Dividing the logarithms instead of subtracting them.

Takeaway log turns $\times$ into $+$ and $\div$ into $-$; combine into one log before evaluating.

Question 16 FUN-15 · Functions and transformations · spec MM8.2

Which of the following is the equation of the graph obtained by stretching $y = x^3 - 3x$ parallel to the x -axis with scale factor $\frac{1}{2}$?

Answer: E

A stretch parallel to the x -axis with factor $\frac{1}{2}$ gives $y = f(2x) = (2x)^3 - 3(2x) = 8x^3 - 6x$.

Full method - how and why

A horizontal stretch of factor $\frac{1}{2}$ replaces x by $2x$ - the reciprocal of the factor.

1 A stretch parallel to the x -axis with scale factor k gives $y = f\left(\frac{x}{k}\right)$; with $k = \frac{1}{2}$ that is $y = f(2x)$.

why: Inside changes act inversely: to squash the graph to half its width you must double the input.

2 Substitute: $y = (2x)^3 - 3(2x)$.

why: Replace every x in the formula, including the one in the linear term.

3 Simplify: $8x^3 - 6x$.

why: $(2x)^3 = 8x^3$ - the coefficient is cubed too.

Where it goes wrong

- Replacing only the first x , giving $8x^3 - 3x$ - an offered option.
- Using $y = f(x/2)$ and getting $\frac{x^3}{8} - \frac{3x}{2}$, which is a stretch of factor 2.

Takeaway Stretch factor k parallel to the x -axis $\Rightarrow$ replace x by x/k ; substitute into **every** occurrence.

Question 17 POL-12 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 3x^2 + 2x + 7$ is divided by $x^2 + 1$, what is the remainder?

Answer: B

$x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x - 3) + (x + 10)$: expanding gives $x^3 - 3x^2 + x - 3 + x + 10$. Remainder $x + 10$.

Full method - how and why

Division by a quadratic again: remainder is linear. Comparing coefficients is fastest because the divisor has no x term.

- 1 Write $x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x + c) + (ax + b)$.
why: Cubic divided by quadratic gives a linear quotient and a linear remainder.
- 2 Expand the right: $x^3 + cx^2 + x + c + ax + b$.
why: $(x^2 + 1)(x + c) = x^3 + cx^2 + x + c$.
- 3 Compare x^2 : $c = -3$.
why: Only the quotient contributes to x^2 .
- 4 Compare x : $1 + a = 2$, so $a = 1$. Compare constants: $c + b = 7$, so $b = 7 + 3 = 10$.
why: Work down the powers; each comparison introduces exactly one new unknown.
- 5 Remainder: $x + 10$.
why: Degree 1 < degree 2 ✓.

Where it goes wrong

- Forgetting the $+x$ that $(x^2 + 1)(x)$ contributes, giving $a = 2$ and the remainder $2x + 10$.
- Using the Remainder Theorem with $x^2 + 1$ - it has no real roots, so that route needs complex numbers and is off-specification.

Takeaway Comparing coefficients handles any divisor; only use root substitution when the divisor has real roots.

Question 18 SIM-16 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $\frac{x-2}{x+1} \leq 0$.

Answer: C

A fraction is ≤ 0 when numerator and denominator have opposite signs (or the numerator is 0): $x - 2 \leq 0$ and $x + 1 > 0$ gives $-1 < x \leq 2$; $x - 2 \geq 0$ and $x + 1 < 0$ is impossible. $x = -1$ is excluded as the fraction is undefined.

Full method - how and why

A rational inequality. Never multiply by $x + 1$ (its sign is unknown) - analyse the sign of the fraction instead.

- 1 Find the critical values: numerator zero at $x = 2$, denominator zero at $x = -1$.
why: The fraction can only change sign where the numerator or denominator vanishes.
- 2 Test the three regions. For $x < -1$ (say $x = -2$): $\frac{-4}{-1} = 4 > 0$. For $-1 < x < 2$ (say $x = 0$): $\frac{-2}{1} = -2 < 0$. For $x > 2$ (say $x = 3$): $\frac{1}{4} > 0$.
why: One test value per region settles the sign throughout that region.
- 3 Include $x = 2$ (the fraction is 0, and ≤ 0 allows it) but exclude $x = -1$ (undefined).
why: The two endpoints behave differently: a zero numerator is allowed by $\leq$; a zero denominator never is.
- 4 Answer: $-1 < x \leq 2$.
why: Note the mixed brackets - strict at -1 , inclusive at 2.

Where it goes wrong

- Multiplying by $x + 1$ to get $x - 2 \leq 0$, i.e. $x \leq 2$, which wrongly includes everything below -1 .
- Writing $-1 \leq x \leq 2$ and including a value where the expression is undefined.

Takeaway For $\frac{A}{B} \leq 0$, use a sign chart on A and B ; a zero denominator is always excluded, a zero numerator is included when the inequality is not strict.

Question 19 DIF-13 · Differentiation · spec MM6.1

$f(x)$ is a polynomial with $f'(x) = (x - 2)^2(x + 1)$. Which of the following is true?

Answer: C

f' changes sign from negative to positive at $x = -1$ (local minimum). At $x = 2$ the factor $(x - 2)^2$ does not change sign, so f' stays positive: a stationary point but not a turning point (a point of inflexion).

Full method - how and why

Reading a factorised derivative. A squared factor does **not** change the sign, so it gives a stationary point that is not a turning point.

- 1 The stationary points are the roots of f' : $x = -1$ and $x = 2$.
why: $f'(x) = 0$ at both.
- 2 Near $x = -1$: $(x - 2)^2 > 0$ always, so the sign of f' is the sign of $(x + 1)$ - negative then positive.
why: A sign change from $-$ to $+$ means the function falls then rises: a local minimum.
- 3 Near $x = 2$: $(x + 1) > 0$ there, and $(x - 2)^2 \geq 0$, so $f' \geq 0$ on both sides - no sign change.
why: A repeated (even-power) factor touches zero without crossing, so the gradient does not change sign.
- 4 Conclusion: local minimum at $x = -1$; stationary but **not** a turning point at $x = 2$.
why: That is a stationary point of inflexion.

Where it goes wrong

- Assuming every root of f' is a turning point.
- Using $f''(2) = 0$ to conclude 'maximum' or 'minimum' - a zero second derivative is inconclusive.

Takeaway Odd-power factors of f' change sign (turning points); even-power factors do not (inflexions).

Question 20 IND-18 · Indices and surds · spec MM1.2

The expression $\frac{3}{7 - 2\sqrt{3}} - \frac{3}{7 + 2\sqrt{3}}$ simplifies to

Answer: F

Common denominator $(7 - 2\sqrt{3})(7 + 2\sqrt{3}) = 49 - 12 = 37$. Numerator: $3(7 + 2\sqrt{3}) - 3(7 - 2\sqrt{3}) = 12\sqrt{3}$.

Full method - how and why

A difference of two rationalisable fractions. Using a single common denominator is faster than rationalising each one.

- 1 Use the common denominator $(7 - 2\sqrt{3})(7 + 2\sqrt{3}) = 49 - 12 = 37$.
why: The two denominators are conjugates, so their product is automatically rational - no separate rationalising needed.
- 2 Numerator: $3(7 + 2\sqrt{3}) - 3(7 - 2\sqrt{3}) = 3 \times 4\sqrt{3} = 12\sqrt{3}$.
why: When you subtract conjugates the rational parts cancel and the surd parts double - so only the surd survives.
- 3 Result: $\frac{12\sqrt{3}}{37}$.
why: Already rationalised, since the denominator is an integer.

Where it goes wrong

- $(2\sqrt{3})^2 = 4 \times 3 = 12$, not 6; using 6 gives denominator 43.
- Adding the numerators instead of subtracting, giving $\frac{42}{37}$ - which is offered.

Takeaway A difference of conjugate fractions leaves only the surd part; their product denominator is rational from the start.